

Pathophysiology of spasticity

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01 Introduction

01 Introduction

Classic Definition (Lance, 1980)

A motor disorder characteristic by **velocity-dependent** increase in tonic stretch reflexes with exaggerated tendon jerks resulting from hyperexcitability of stretch reflex as one component of UMN syndromes

Modern Definition (Li et al., 2021)

Velocity and muscle length-dependent increase in resistance to externally imposed stretch.

Terminology

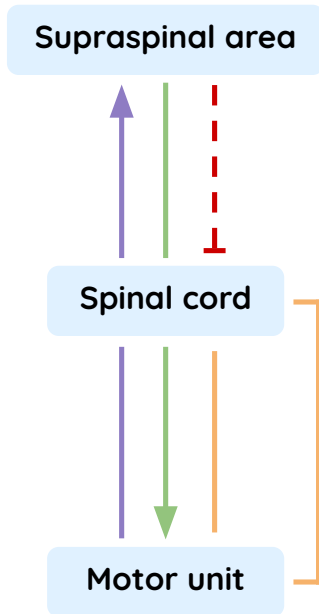
While "spasticity" is frequently used clinically as a term for any muscle tightness following a central nervous system lesion, it is one **specific positive feature** of the Upper Motor Neuron Syndrome (UMNS)

Upper Motor Neuron Syndrome	
Positive component (Muscle overactivity)	Negative component (Loss of function)
• Spasticity (Velocity-dependent) • Dystonia (Sustained) • Hyperreflexia • Release of primitive reflexes	• Muscle Weakness / Paralysis • Loss of dexterity / Coordination • Fatigue • Flaccidity (Initial Phase)

	Classic Definition (Lance, 1980)	Modern Definition (Li et al., 2021)
Primary Mechanism	Spinal reflex arc hyperexcitability alone.	Hyperexcitable descending excitatory brainstem pathways (reticulospinal) drive elevated spinal motor neuron excitability.
Associated Motor Features	Focuses narrowly on the isolated stretch reflex.	Recognizes coexisting abnormal synergies, inappropriate muscle activation, and anomalous co-activation sharing similar pathophysiological origins.
Central vs. Peripheral Scope	Reserves "spasticity" for alterations of central processing of sensory input, excluding structural muscle changes.	Acknowledges that clinical spasticity interacts dynamically with intramuscular and extracellular matrix adaptations.

02 Normal motor function

02 Normal motor function



1. **Sensory Feedback Integration:**
Real-time processing of muscle length, stretch velocity, muscle tension, and joint position.
2. **Reflex Circuit Regulation:**
Supraspinal suppression to prevent spinal reflex arcs from becoming hyperexcitable.
3. **Volitional Movement Coordination:**
Precise agonist activation paired with antagonist relaxation (reciprocal inhibition).

Motor unit

is the **Final Common Pathway** of the motor system

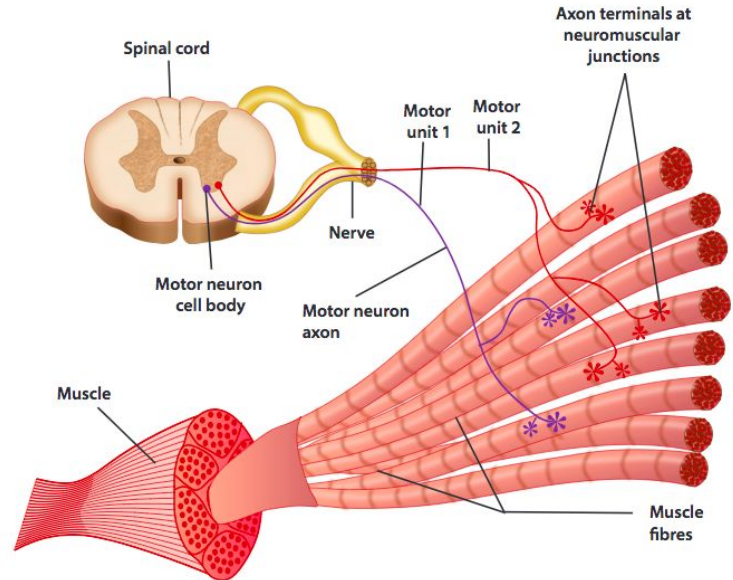
EXTRAFUSAL FIBERS

α -motor neuron

+

Muscle fibers

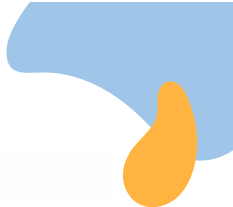
(All extrafusal muscle fibers that are innervated by it)



Muscle fiber



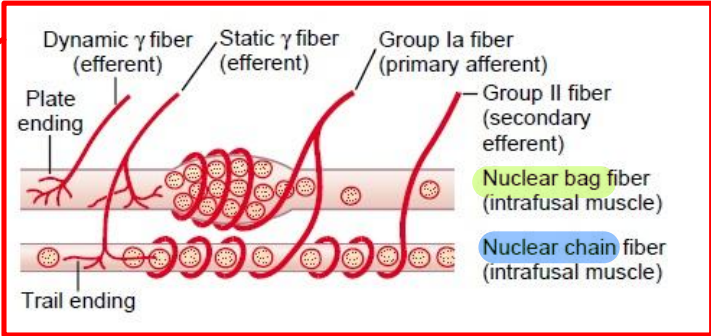
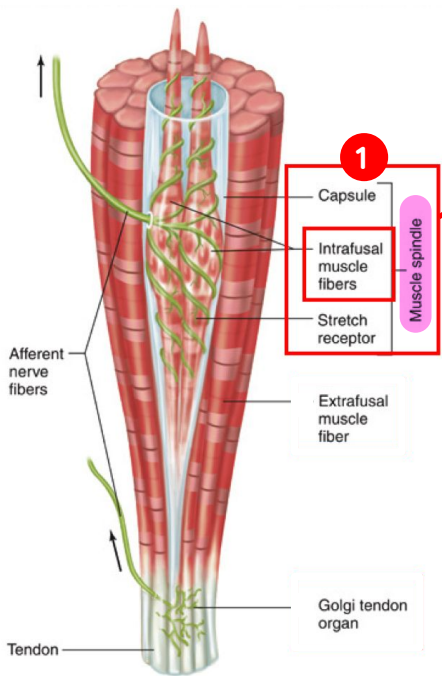
Property	Type I Fibers (Slow-Twitch)	Type II Fibers (Fast-Twitch)
Biological Profile	Small diameter, red , oxidative, fatigue-resistant	Large diameter, white , anaerobic, high power output
Primary Normal Function	Maintains baseline tonic posture and low-velocity control	Generates rapid force, power, and acceleration
Recruitment Order	Recruited first	Recruited later for higher force or high-velocity tasks



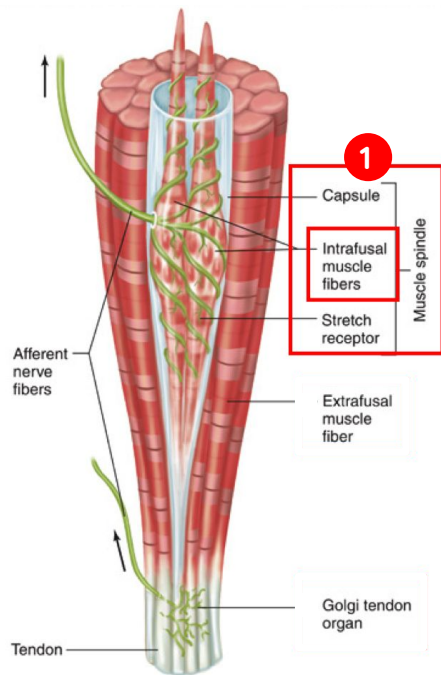
Skeletal Muscle Fiber Characteristics

	Type 1 (Slow Oxidative)	Type 2A (Fast Oxidative Glycolytic)	Type 2B (Fast Glycolytic)
Major source of ATP	Oxidative phosphorylation	Oxidative phosphorylation	Glycolysis
Mitochondria	High	High	Low
Myoglobin content	High	High	Low
Capillarity	High	High	Low
Muscle color	Red	Red	White
Glycogen content	Low	Intermediate	High
Glycolytic enzyme activity	Low	Intermediate	High
Myosin ATPase activity	Low	High	High
Speed of contraction	Slow	Fast	Fast
Rate of fatigue	Slow	Intermediate	Fast
Muscle fiber diameter	Small	Intermediate	Large

Spindle fiber

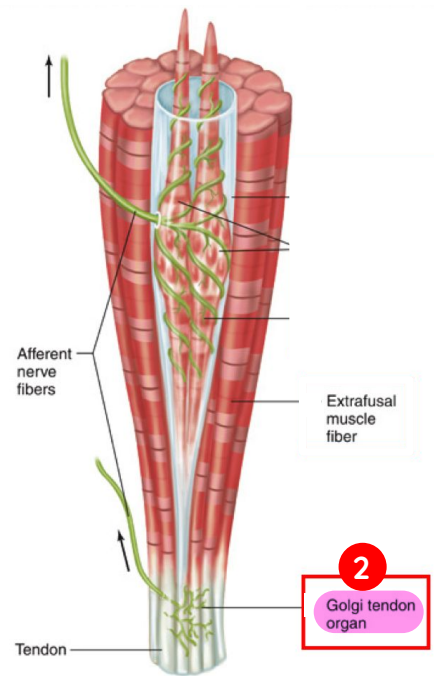


Nuclear bag	Nuclear chain
Ia afferent fiber	Ia, II afferent fiber
<u>Rate of change</u> in muscle length	<u>Steady state</u> of muscle length
dynamic stretch	static stretch



	Muscle Spindle
Anatomic Orientation	Attached in parallel with extrafusal muscle fibers.
Parameter Sensed	Muscle Length and Rate of Stretch (Velocity) .
Sensory Afferents	<i>in nuclear bag</i> Type Ia (velocity & length) and <i>in nuclear chain</i> Type II (static length).
Motor Innervation	<i>by Extrafusal α-motor neuron</i> γ-Motor Neurons (adjusts intrafusal fiber tension).
Primary Reflex Circuit	Monosynaptic Stretch Reflex Arc : Excites agonist α -motor neuron directly.
Effect on Agonist	Facilitates/contracts agonist muscle.
Effect on Antagonist	Inhibits antagonist via Ia interneuron (Reciprocal Inhibition)
Functional Role	Maintains muscle tone, posture, and length feedback throughout ROM.

GTO

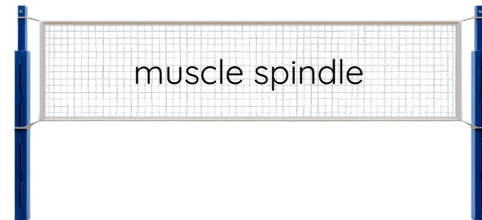
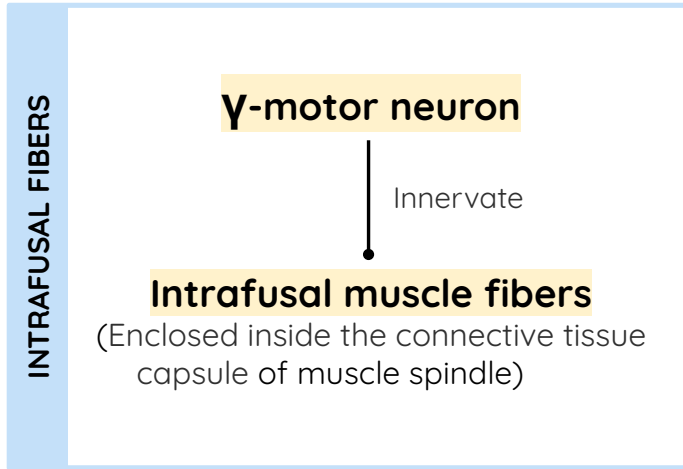


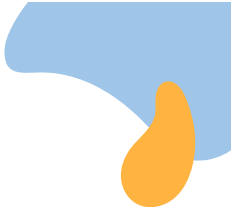
	Golgi Tendon Organ (GTO)
Anatomic Orientation	Attached in series with extrafusal fibers at the musculotendinous junction.
Parameter Sensed	Muscle Tension (Contractile Force).
Sensory Afferents	Type Ib afferents.
Motor Innervation	<u>None</u> (purely a sensory receptor).
Primary Reflex Circuit	Polysynaptic Autogenic Inhibition: Acts via Ib interneurons.
Effect on Agonist	Inhibits/relaxes the agonist muscle.
Effect on Antagonist	Facilitates/contracts antagonist via Ib interneuron.
Functional Role	Imposes a ceiling effect on maximum force to prevent tendon tear.

α - γ Coactivation

During voluntary movement, γ -motor neuron fires simultaneously with the α -motor neuron.

Purpose : Prevents muscle spindle from slacking when the main muscle contracts, preserving sensitivity throughout the full range of motion (ROM).





Spinal (segmental) circuit

Spinal interneuron

1. Stretch reflex arc

Excitatory pathway

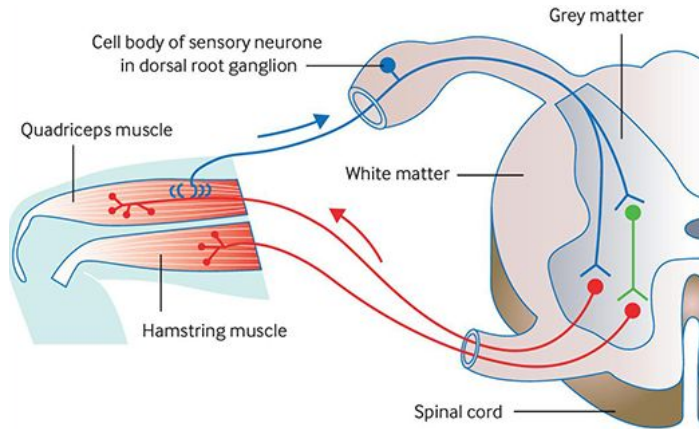
2. Reciprocal inhibition

3. Autogenic inhibition (Ceiling effect)

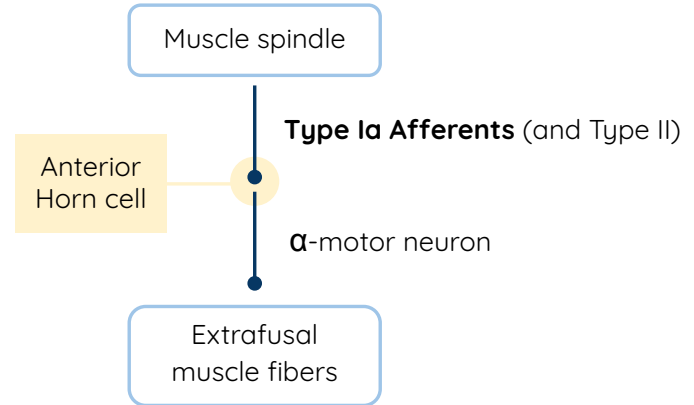
4. Renshaw cell (Recurrent inhibition)

Inhibitory pathway

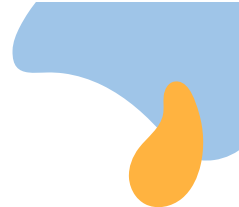
Stretch reflex arc



Changes in muscle length
and rate of stretch



Contracts the agonist muscle
to oppose the applied stretch



Stretch reflex arc

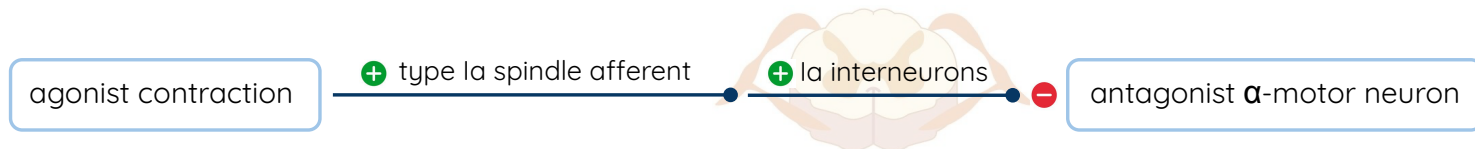
	Phasic Stretch Reflex (Dynamic)	Tonic Stretch Reflex (Static)
Trigger	Rapid, sudden stretch (e.g., reflex hammer tap, quick passive movement).	Sustained, continuous stretch (e.g., baseline posture, slow muscle lengthening).
Primary Receptor	Type Ia Afferents (sensory to rate of change).	Type II Afferents & Type Ia (sensory to static length).
Clinical Manifestation	Deep Tendon Reflexes (DTRs) and the velocity-dependent "Catch" in spasticity.	Baseline muscle tone and resistance felt throughout passive range of motion.

Reciprocal inhibition

→ - မိသားစု Agonist ၊ သို့မဟုတ် relax Antagonist

Mechanism: Triggered by type Ia spindle afferents during agonist contraction. Ia interneurons **inhibit the antagonist motor neuron**, allowing the antagonist to relax smoothly.

Requires **supraspinal facilitation** from descending pathways to maintain effective reciprocal inhibition.



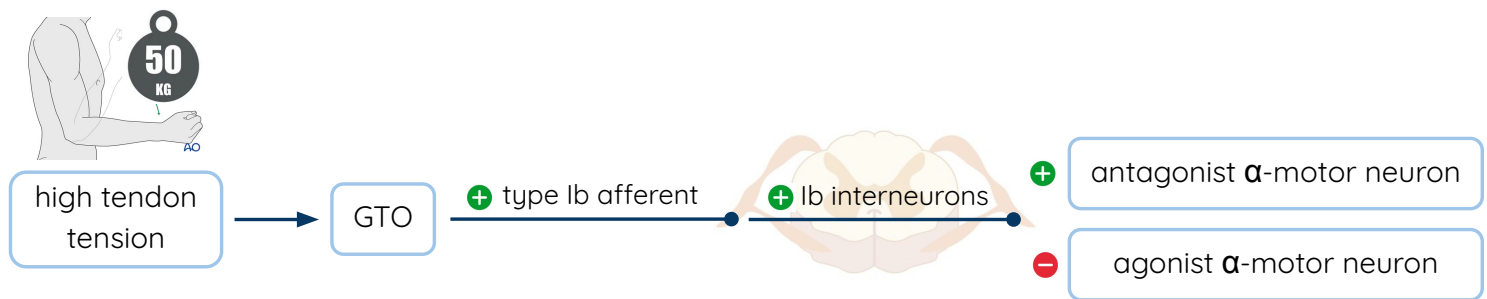
Autogenic inhibition (Ceiling effect)

သို့လည်းကောင်း muscle tendon သို့လည်းကောင်း
if သို့လည်းကောင်း Golgi tendon အား

Mechanism: Triggered by **type Ib** afferents from **Golgi tendon organ** (GTO) during high tension. Ib interneurons **inhibit agonist motor neuron** and **facilitate the antagonist**.

- 1) relax agonist
- 2) contract antagonist

Function: Establishes a **ceiling effect** on maximum force generation to prevent musculotendinous avulsion or tear.

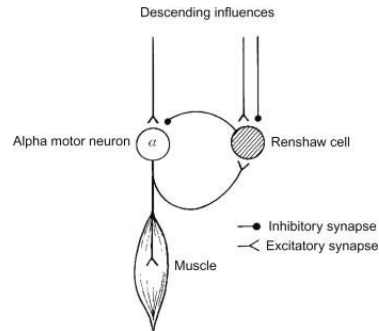


Renshaw cell (Recurrent inhibition)

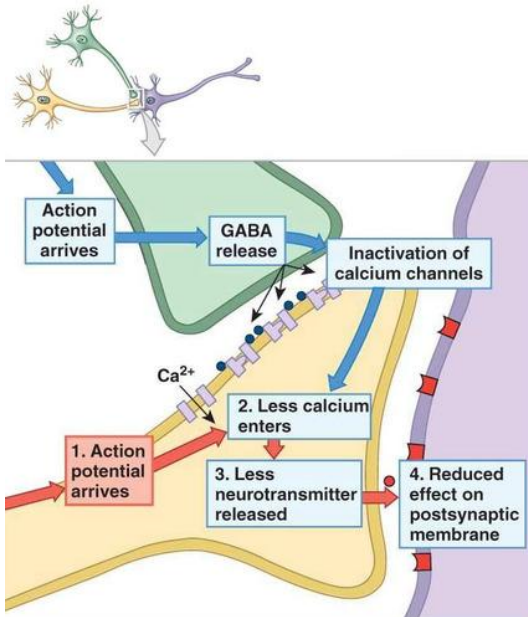
locate in spinal cord

Mechanism: Activated by recurrent collateral axon branches from α -motor neuron itself.

Function: Sends a **negative feedback signal** to shut off or damp excessive, repetitive firing of the originating α -motor neuron while facilitating antagonist Ia interneurons, ensuring smooth motor execution.



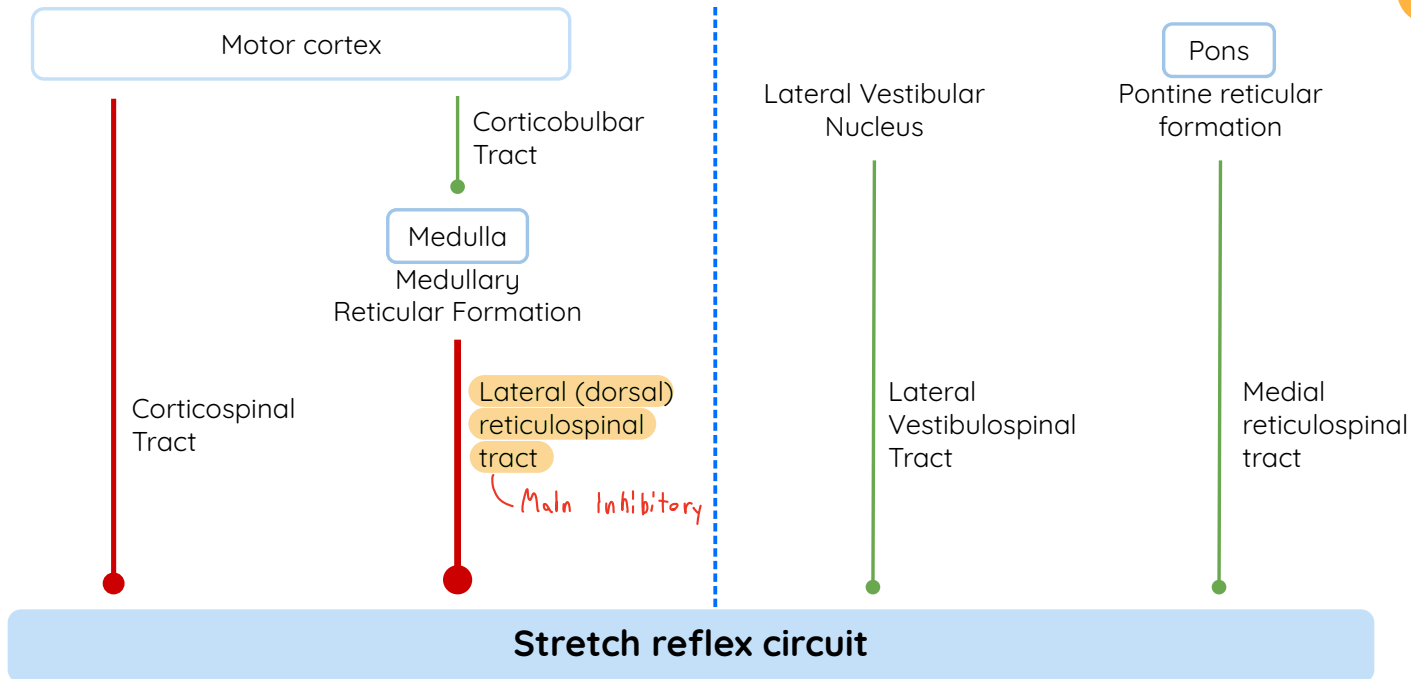
Presynaptic inhibition (via GABA)

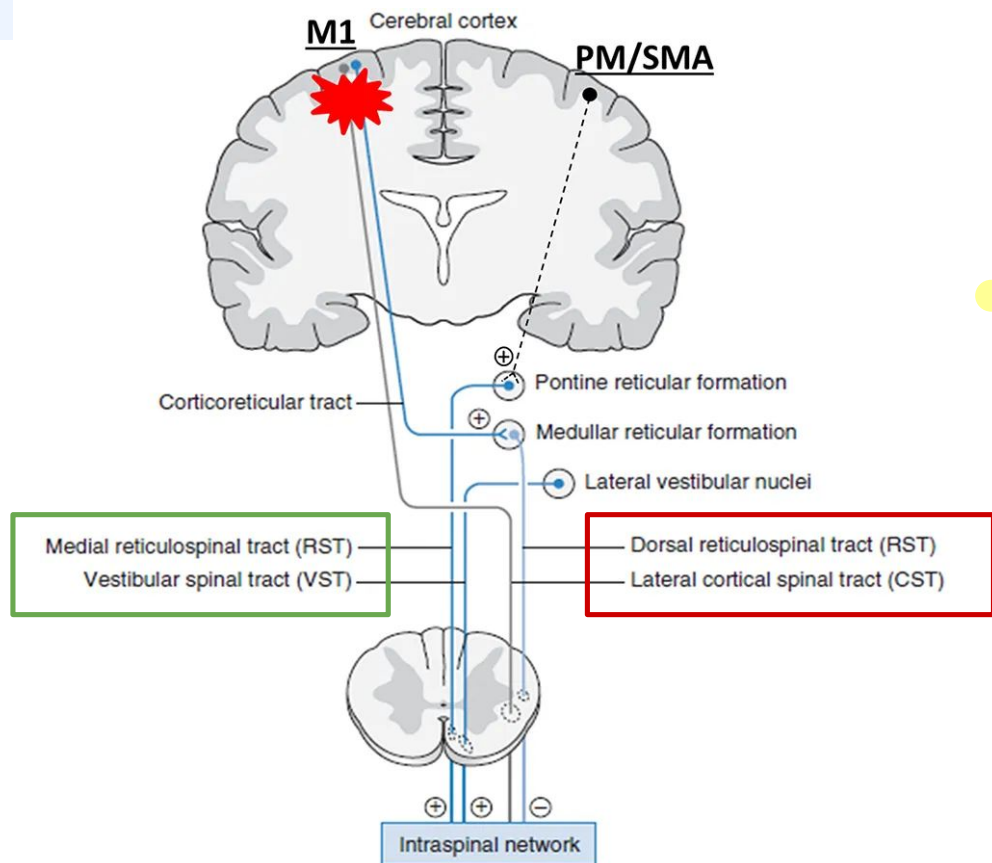


- Presynaptic inhibition neurons are controlled by descending inhibition pathway.
- GABA binds to Ca²⁺ channel in presynaptic neuron
- > inhibit glutamate release to postsynaptic neuron
- > decrease AP

Descending pathways

● = excitatory
● = inhibitory





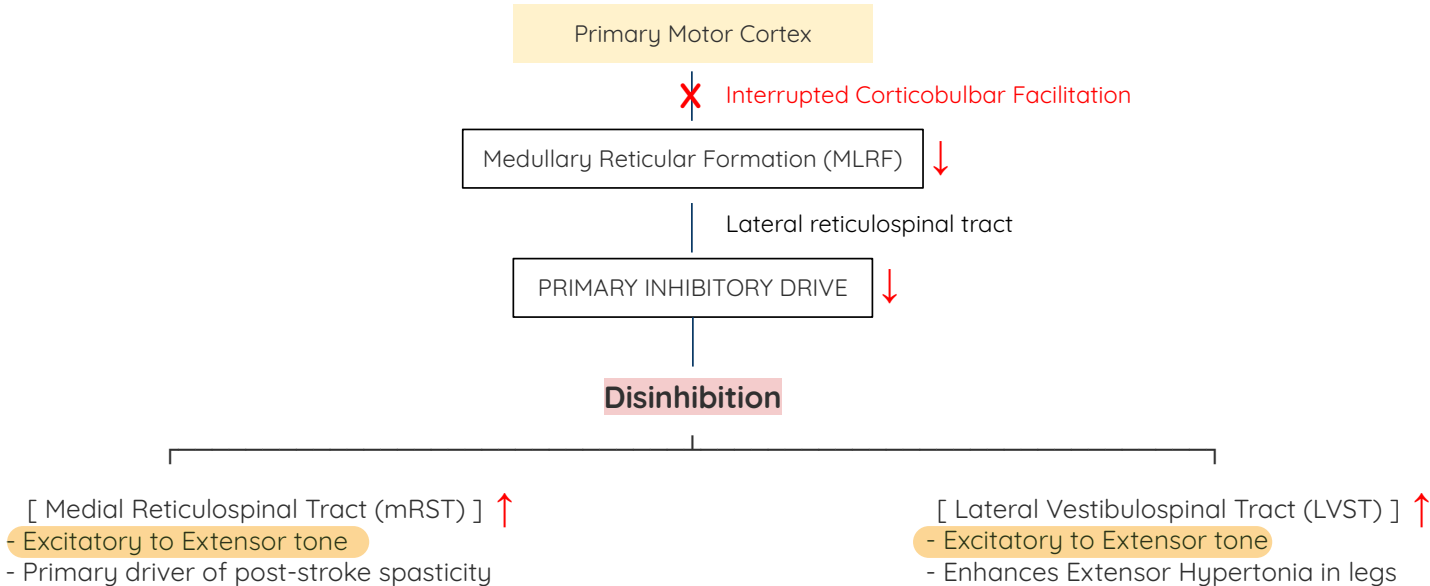
Descending Pathway	Anatomical Origin	Normal Physiological Function	Impact of UMN Lesion (Disinhibition)
Corticospinal Tract (CST)	Primary motor cortex	Controls volitional, fine coordinated motor tasks (finger dexterity).	Primary cause of weakness and loss of dexterity (negative signs) rather than spasticity.
Lateral (dorsal) reticulospinal tract (LRST)	Medulla (lower brainstem).	Primary Inhibitory Center: Suppresses extensor muscle tone and stretch reflex arcs.	Loss of cortical facilitation to MLRF leads to disinhibition, allowing excitatory pathways to run unchecked.
Medial Reticulospinal Tract (MRST)	Pons (mid-brainstem).	Excitatory Pathway: Facilitates extensor α and γ -motor neurons.	Becomes hyperexcitable and unopposed, serving as the primary driver of post-stroke spastic hypertonia.
Lateral Vestibulospinal Tract (LVST)	Lateral Vestibular Nucleus (Ventromedial cord).	Excitatory Pathway: Facilitates extensor tone and inhibits flexors.	Provides unopposed extensor drive, exacerbating lower-extremity extensor hypertonia.
Corticorubrospinal / Rubrospinal Tract	Red Nucleus / Midbrain.	Flexor Facilitation: Excites flexor motor neurons and inhibits extensors via interneurons.	Loss of coordinated flexor modulation.

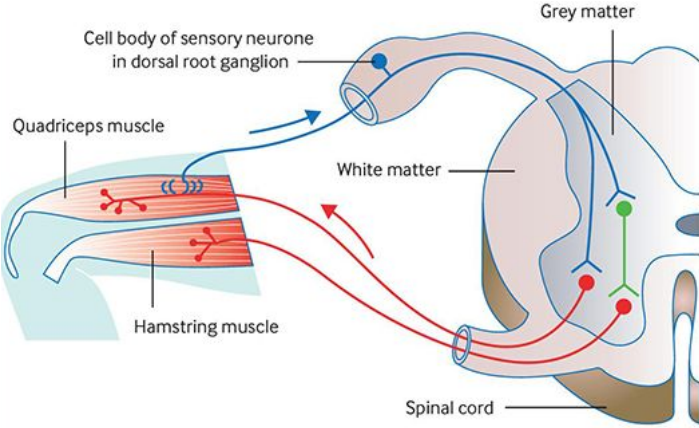
03 Pathophysiology

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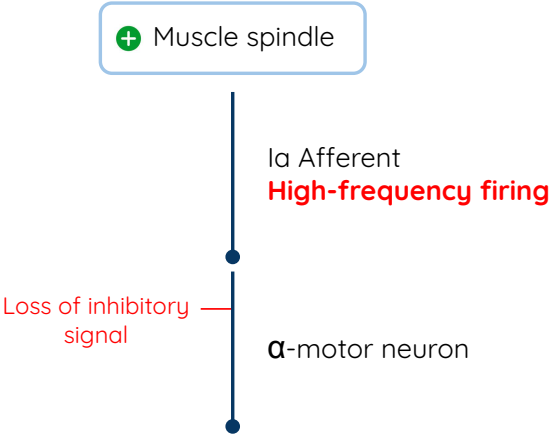
Supraspinal lesion

Stroke, TBI





Changes in muscle length and rate of stretch (Quick stretching)



Exaggerated Monosynaptic Response

1

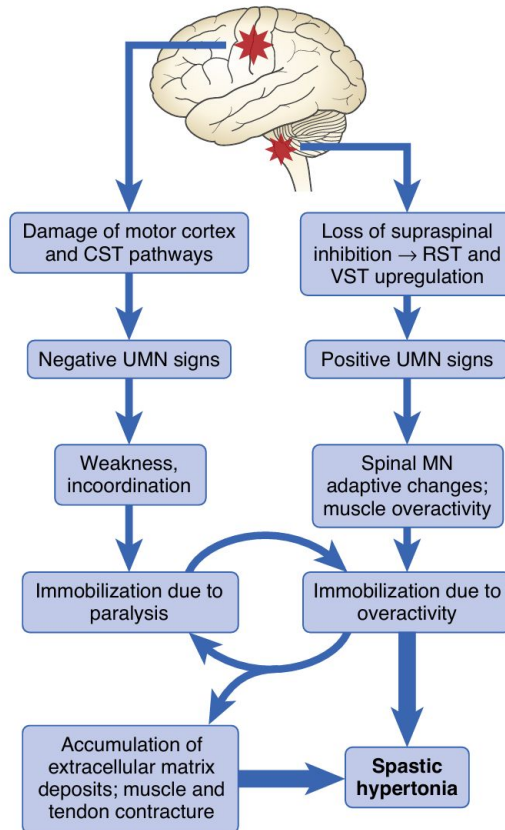
Supraspinal lesion

Stroke, TBI

Disinhibition Mechanism: The Medullary Lateral Reticular Formation acts as the primary central **brake** on extensor muscle tone and stretch reflexes. Because MLRF requires continuous excitatory input from the motor cortex via corticobulbar projections, cortical damage shuts off this inhibitory pathway.

MRST Hyperexcitability: Without MLRF inhibition, **Medial Reticulospinal Tract (MRST)** and **Lateral Vestibulospinal Tract (LVST)** operate unchecked. Unopposed mRST firing into spinal stretch reflex circuits elevates the excitability of alpha-motor neuron pools, acting as the primary neural driver of post-stroke spasticity.

Corticospinal Tract (CST) Role: Direct CST damage primarily accounts for **weakness and loss of fine motor dexterity** (negative signs) rather than spasticity itself



Spastic hypertonia mechanisms

Subcortical areas/brainstem

- Reticulospinal upregulations
- Vestibulospinal upregulations
- Rubrospinal upregulations

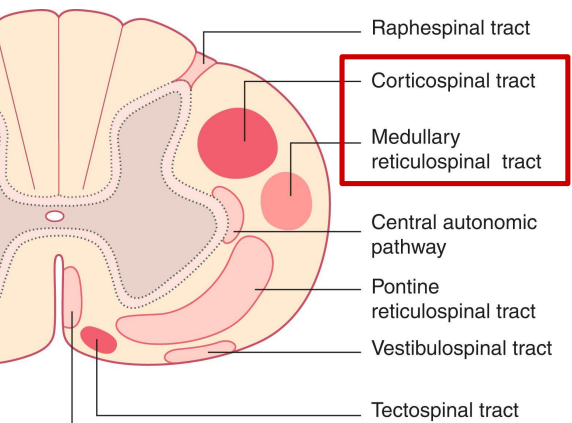
Spinal MNs

- (secondary and adaptive changes)
- Motor neuron hyperexcitability
 - Hyperexcitable for the remaining stretch reflex arc

Muscles

- (secondary and adaptive changes)
- Increased mechanical stiffness

2 Spinal lesion



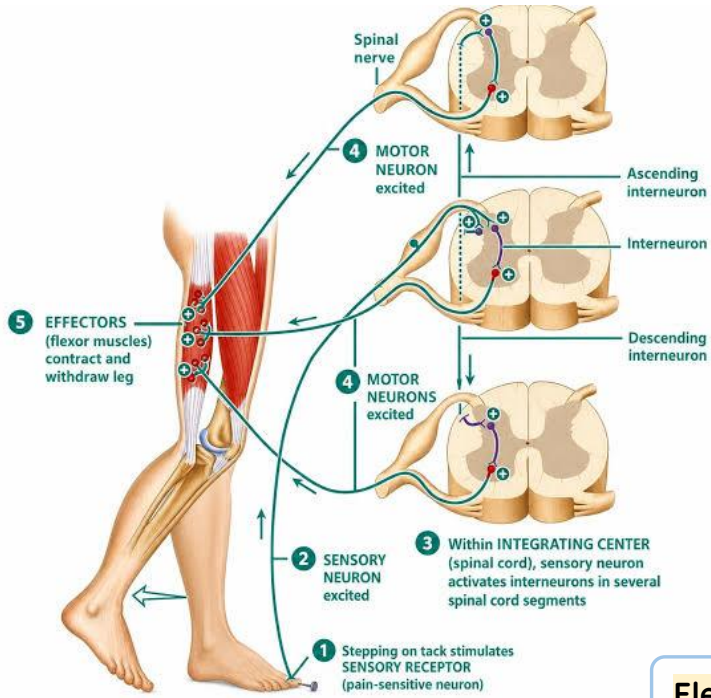
Incomplete SCI

✗ Damage to Dorsolateral Cord
(CST + Lateral reticulospinal tract)

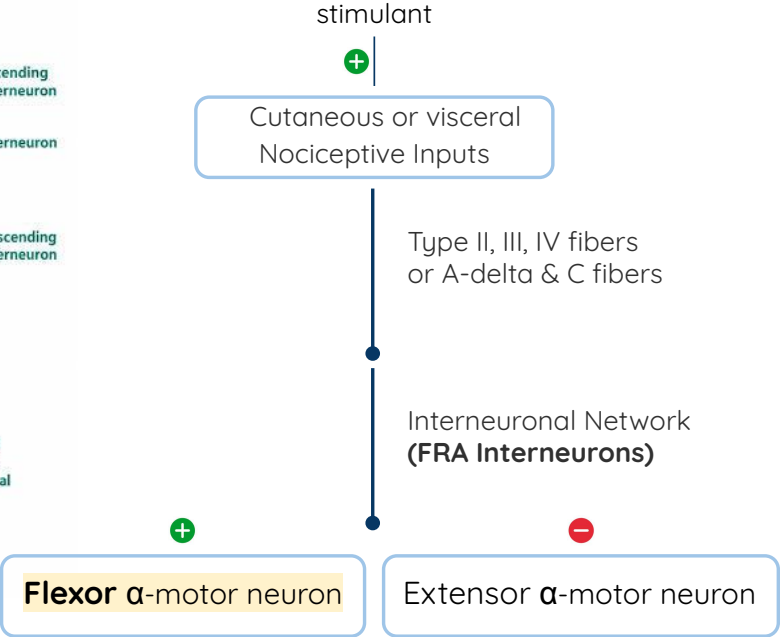
Loss of inhibitory LRST

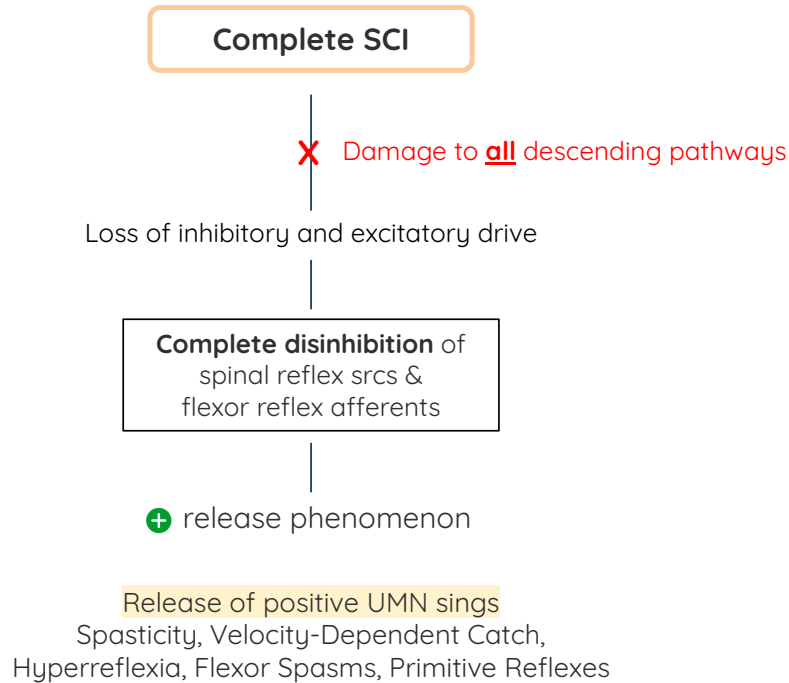
Unopposed Excitatory drives
(MRST + LVST) ↑

↓
flexor spasm, hyperreflexia, and spasticity



Flexor Reflex Afferents





Acute phase: Spinal shock

(0 – 6 weeks)

- Acute loss of supraspinal drive
- Motor neuron pool in hyperpolarized state
- Flaccidity & Areflexia

Maladaptive
Neuroplasticity



Chronic phase: Spinal plasticity

(After 6 weeks)

- Complete disinhibition
- Sprouting & receptor upregulation
- Severe Spasticity & Flexor reflex spasms

3 Neurotransmitter mechanism

Inhibitory neurotransmitter

GABA: Main inhibitor in synapse;
Impaired synthesis of GABA in spinal inhibitory interneurons

Glycine: Postsynaptic inhibition;
Reduced glycinergic function contributes to hyperexcitability

Dopamine: Both excitatory and inhibitory roles;
Altered dopaminergic modulation in CNS > uncleared mechanism

3 Neurotransmitter mechanism

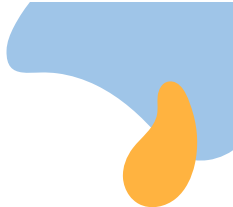
Excitatory neurotransmitter

Glutamate: Main excitatory in synapse; Not necessarily increased, but reduced inhibitory counterbalance and up-regulation of its receptors

Norepinephrine: Both excitatory and inhibitory roles; Altered noradrenergic modulation in CNS

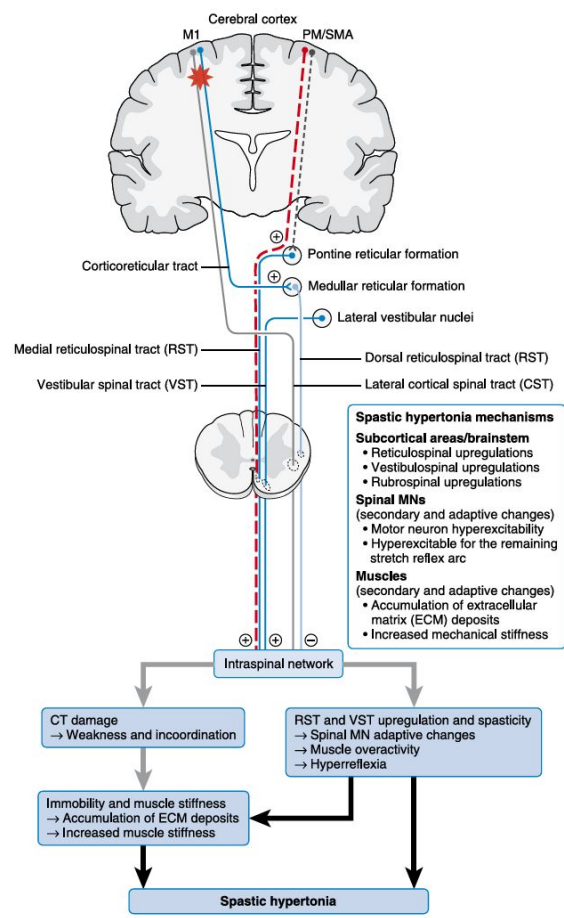
Acetylcholine: Contribute to spinal circuit function;

Substance P: Excitatory in spinal cord and motor reflexes; Excitability of spinal circuits, especially related to noxious stimuli



Sheean's Categories of Muscle Overactivity

Category	Pathophysiological Mechanism	Clinical Features
1. Spinal Reflexes (Afferent-Dependent)	Abnormal spinal processing of peripheral sensory inputs.	• Tonic & phasic stretch reflexes (Spasticity, DTRs) • Clonus, Flexor reflex afferent spasms • Cutaneous reflexes (Babinski sign)
2. Efferent Drives (Afferent-Independent)	Driven directly by supraspinal efferent commands to motor neurons without requiring sensory feedback.	• Spastic Dystonia: Sustained resting posture independent of stretch (e.g., flexed arm posture post-stroke) • Associated Reactions: Involuntary synkinesis triggered during physical effort
3. Voluntary Movement Disorders	Disruption of reciprocal inhibition interneuronal circuits.	• Impaired Reciprocal Inhibition: Antagonist fails to relax, causing pathological co-contraction.



04 Clinical assessment

History taking

Problem identification

- How often? Is the limb tight all the time?
- Which muscles/body part involved? Upper vs lower?
- Are the spasm beneficial/detrimental?
- Does it cause pain?, Have any skin compromises due to tightness/spasm?,
- Does it limit ability to move limbs?
- Are the spasm more frequent/severe than usual?
- Dose particular position trigger tightness?, Is the tightness related to spasms?
- What elicits the spasm?

Previously treatment and the outcome

- Current medication?
- Recent medical problem?

ADVANTAGE

- Aid to standing & walking via extensor spasm
- More isometric strength via facilitation reflex
- Maintained muscle mass
- Maintained bone mineralization
- Reduced dependent edema
- Reduced DVT

ADVANTAGE

- Interfere perineal hygiene & sexual function
- Interfere transfer/standing/ambulation : clonus, scissoring
- Increase skin shearing: pressure sore
- Increase risk of contracture
- Impaired sleep: spontaneous spasm, myoclonus jerk
- Pain: continuous flexor spasm

Physical examination

Observe

- Limb posture at rest and with variation in position

Active ROM

- Pain, discomfort
- Functional length, coordination, spastic co-contraction, contraction

Passive ROM

- Strength, muscle tone, velocity-dependent "angle of catch", clonus, pain
- Spasticity, rigidity, contraction

Functional assessment

- Gait characteristic, postural abnormalities, performance of specific test and task
- Impact of multiple impairments on performance

Physical examination

Direct measurement

stretch reflex

Modified Ashworth scale (MAS)

Tardieu scale

Biomechanic: Pendulum test

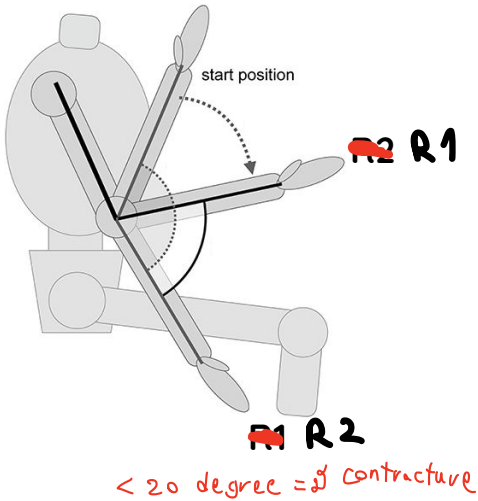
Indirect measurement

Pain Severity, Physical rating scale, Spastic associated reaction, ADL, social activity, Gait analysis/dynamic EMG

Modified Ashworth scale (MAS)

Modified Ashworth Scale	
Grade	Description
0	No increase in muscle tone
1	Slight increase in muscle tone, manifested by a catch and release or by minimal resistance at the end of the range of motion when the affected part(s) is moved in flexion or extension
1+	Slight increase in muscle tone, manifested by a catch, followed by minimal resistance throughout the remainder (less than half) of the ROM
2	More marked increase in muscle tone through most of the ROM, but affected part(s) easily moved
3	Considerable increase in muscle tone, passive movement difficult
4	Affected part(s) rigid in flexion or extension

Tardieu scale



R2-R1 > 20 degrees
= Spasticity

Tardieu	
Velocity to Stretch	
V1	As slow as possible
V2	Speed of the limb segment falling (with gravitational pull)
V3	At a fast rate (>gravitational pull)
Quality of Muscle Reaction	
0	No resistance throughout passive movement
1	Slight resistance throughout, with no clear catch at a precise angle.
2	Clear catch at a precise angle followed by release
3	Fatiguable Clonus (< 10 secs) occurring at a precise angle
4	Unfatiguable Clonus (> 10 secs) occurring at a precise angle
5	Joint immobile
Spasticity Angle	
R1	Angle of catch seen at Velocity V2 or V3
R2	Full range of motion achieved when muscle is at rest and tested at V1 velocity

References

- Goetz L. Stiens S. Spasticity. In: **Braddom's Physical medicine and rehabilitation. Seventh edition.** Saunders/Elsevier: 2025. 485-507.
- Stitik T. Klecz R., Greenwald B., et al. Spasticity and Muscle Overactivity as Components of the Upper Motor Neuron Syndrome. In: **DeLisa's physical medicine and rehabilitation: principles and practice. Fifth edition.** Philadelphia. Wolters Kluwer; 2010. 1319-1337



Thank you